NASA SPACE APPS CHALLENGE 2026 · FINAL COMPETITION RELEASE

MITOCHONDRIA

NASA Evaluator Quick Start Guide

FINAL · 23 October 2026 · a 15–20-minute guided demonstration

https://mitochondria.edocinfo.com

A functional, integrated, open-source astronaut health intelligence research platform

Research prototype with human oversight. Synthetic data only. Not NASA-endorsed, not clinically validated, not flight-qualified, not connected to NASA astronaut telemetry.

1. Purpose and Duration

This guide lets an evaluator see MITOCHONDRIA working end to end in **15–20 minutes** — without administrative database access, without real patient data and without any mandatory physical sensor. It shows a functional, integrated research platform: live monitoring that never invents a value, offline-first encrypted storage, mission-scoped roles enforced in the database, synthetic laboratory-evidence review with independent Ground Medical oversight, and research-only scientific interpretation, including the Health Intelligence Research Explorer of 54 provisional biomarker proposals.

Table 1. Demonstration paths

Path	Duration	Needs
A · Public tour	≈ 8 minutes	A browser. No account.
B · Role-based workflows	≈ 11–12 minutes	Synthetic role accounts provided by the project owner. Every role step has an alternative if an account is not available.

Live application	https://mitochondria.edocinfo.com
Companion documents	Master Operations & Technical Manual (FINAL), the 30-slide competition presentation (FINAL) and the Final Competition Release Report
Status labels	OPERATIONAL · FIELD-VERIFIED · SYNTHETIC DEMONSTRATION · RESEARCH-ONLY · VALIDATION BOUNDARY · OUT OF SCOPE — defined in Manual Section 1.2

NOTE — Some pages show a short module label above the page title (for example “P17 · Simulation & synthetic mission dataset”). These labels identify the software module and can be ignored during the demonstration.

2. What You Will and Will Not See

You will see	You will not see — by design
Mission-scoped roles and identity states	Diagnoses, risk scores, readiness verdicts, alerts or treatment
Live capture states with honest freshness and storage evidence	Fabricated sensor values — unconnected cards stay DISCONNECTED
Local, encrypted, offline-first operation	Live NASA astronaut data
Synthetic laboratory-report review with exact catalogue matching	Real medical documents
Independent Ground Medical review of synthetic batches	Clinical approval — research decisions are synthetic and research-only
Research-only science: handbooks, draft catalogues, NASA OSDR metadata	Activated scientific rules — scientific content awaits independent review

3. Before You Start

Table 2. Prerequisites

Requirement	Detail
Device	Laptop or desktop with a current Chrome, Edge, Firefox or Safari. Optional: an Android phone with Chrome for the camera pulse step.
Connection	HTTPS access to the live application. One optional step uses airplane mode.
Accounts (Path B)	Synthetic Crew Self, Mission Operator and Ground Medical accounts issued by the project owner through an approved channel. This guide publishes no credentials. Use a separate browser profile or private window per role.
Mission	The synthetic mission the owner has assigned to your accounts.
Synthetic file	The repository's synthetic laboratory-report fixture (TXT gives the most reliable exact matching). Never use a real report.
Ground rules	Use only the access your account has; never attempt to bypass role-based access. Use create, save-to-Cloud or decision buttons only when the owner has authorized that action for your session.

3.1 Exact URLs used in the demonstration

Table 3. Route reference (… = https://mitochondria.edocinfo.com)

Page	URL	Account
Home	https://mitochondria.edocinfo.com/	None
Device Integration Center	https://mitochondria.edocinfo.com/mission/devices	None
Live Monitoring	https://mitochondria.edocinfo.com/mission/live	None (Crew Self for mission-bound capture)
Mission Scenario	https://mitochondria.edocinfo.com/mission/simulation	None
NASA Scientific Evidence	https://mitochondria.edocinfo.com/mission/nasa-evidence	None
Health Intelligence + Research Explorer	https://mitochondria.edocinfo.com/mission/intelligence	None (Crew Self for private evidence)
Validation & Usability	https://mitochondria.edocinfo.com/mission/validation	None
Challenge Demo & Submission	https://mitochondria.edocinfo.com/mission/submission	None
Mission Identity & Sign-in	https://mitochondria.edocinfo.com/mission/identity	Owner-provided synthetic account
Mission Workspace · Guidance · Documents	…/mission/experience · …/mission/guidance · …/mission/documents	Crew Self
Domain pages	…/mission/immune-health · …/mission/cardiovascular-health · …/mission/bone-health · …/mission/behavioral-health	Reference public; evidence Crew Self
Alerts · Offline · Records	…/mission/alerts · …/mission/offline · …/mission/records	Crew Self
Mission & Crew Operations	https://mitochondria.edocinfo.com/mission/operations	Mission Operator
Ground & Research Console	https://mitochondria.edocinfo.com/mission/ground	Ground Medical
Source code	https://github.com/computech-gr/astro-vitality-core	None (public repository)

4. Sign-in and Mission Selection

1. Open https://mitochondria.edocinfo.com → **Mission Console → Mission Identity & Sign-in** (`/mission/identity`).
2. Enter the e-mail and password of the synthetic account for the role you are demonstrating and select **Sign in**. A wrong password produces the same generic message for every failure, “Incorrect email or password. Please try again.”, which prevents account enumeration.
3. Confirm **Identity: AUTHENTICATED** and that **Active memberships** is at least 1.
4. Select the membership card for your synthetic mission and role. It becomes **ACTIVE**.
5. For Crew Self, confirm **Live sensor context: MISSION-BOUND**.
6. If no membership appears, the account has no mission access yet — use the alternative path for that role (Section 6).

5. Demonstration Sequence

Times are cumulative and approximate. EVIDENCE is what you can point to on screen. All values are synthetic or absent; no reading is ever invented.

Table 4. Demonstration steps

STEP	TIME	ROLE	SCREEN	ACTION	EXPECTED RESULT	EVIDENCE	FAILURE RECOVERY
A1	0:00–1:00	Public	Home `/`	Read the hero, health loop and capability cards; open a header menu and global search (Ctrl/⌘+K)	Home plus five menus and Search; Apache-2.0 notice in the footer	Header and footer	Reload; check HTTPS
A2	1:00–2:00	Public	Device Integration Center `/mission/devices`	Read the three capability cards	Badges AVAILABLE / USER PERMISSION / UNAVAILABLE HERE for this browser	Capability badges	Bluetooth unavailable on iPhone is expected
A3	2:00–3:30	Public (optional phone)	Live Monitoring `/mission/live`	Observe the mission context and readiness panel; optionally Start camera pulse with a fingertip on the lens, then Stop camera monitoring	No values without a source (cards DISCONNECTED). With a phone: Heart rate LIVE; received samples and confirmed encrypted writes increase; nothing after Stop	Readiness counters; optional non-value receipt	Skip the sensor — physical capture is optional. Permission denied → allow the camera or skip
A4	3:30–4:30	Public	Mission Scenario `/mission/simulation`	Choose Mission Day 127 (seed 1272026); select Regenerate same seed	Identical fingerprint; every observation tagged synthetic	Fingerprint value	—
A5	4:30–5:30	Public	NASA Scientific Evidence `/mission/nasa-evidence`	Enter OSD-48 → Preview dataset → Import research sample metadata	Bounded metadata table labelled research-only; no crew binding	JSON labels research_only, subjectBinding:none	“NASA API error” means the NASA service is unavailable; continue
A6	5:30–7:00	Public	Health Intelligence `/mission/intelligence` — section “Scientific mapping · research review”	Type WBC in “Search by source or candidate”; clear it; choose “Potential equivalence verify protocol”; return to All classifications and select Show 12 more	WBC → 1 proposal; the filter → 11; paging → 24 rendered of 54; the status line always reads 0 independently approved · 0 operationally activated	Status line; proposal cards with rationale and source unit	Section missing → reload; the explorer is research-only metadata and needs no account

STEP	TIME	ROLE	SCREEN	ACTION	EXPECTED RESULT	EVIDENCE	FAILURE RECOVERY
A7	7:00–8:00	Public	Validation `/mission/validation` and Challenge Submission `/mission/submission`	Read the validation criteria and the submission evidence status	Validation criteria with counts; independent scientific validation shown as not yet approved; submission card lists ready and pending items, never converting pending to pass	On-screen status cards	—
B1	8:00–9:00	Crew Self	Mission Identity `/mission/identity`	Sign in; select the Crew Self card	AUTHENTICATED; ACTIVE card; MISSION-BOUND	Identity summary	No card → ALT-CS
B2	9:00–9:30	Crew Self	Mission Workspace `/mission/experience`	Read the task cards	Crew Self cards, including Self-assessment and guidance	Card set	“No tasks available…” → re-select the membership
B3	9:30–10:15	Crew Self	Action Guidance `/mission/guidance`	Answer the six questions → Review protocol categories → Clear session	“Your review — not a medical assessment”; categories with protocol ID and version; nothing stored	On-screen result	Not Crew Self → read the protocol library
B4	10:15–12:30	Crew Self	Mission Documents `/mission/documents`	Tick consent → Upload Laboratory Report (synthetic TXT) → Select All → observation date → manual confirmation → open the local research preview. Save to Cloud only if the owner has enabled and authorized it	Only exact name + unit matches; reference ranges skipped; routing preview by domain; Cloud control shows either Save (n) or pending activation	Candidate table; routing preview; preview table	No matches → use the TXT fixture; disabled controls → check Crew Self context and consent
B5	12:30–13:30	Crew Self	Immune Changes `/mission/immune-health` and Cardiovascular `/mission/cardiovascular-health`	Search the immune proposal catalogue; expand the cardiovascular handbook; tick “Also show original Greek…”	35 immune proposals marked pending independent review and not active; 96 cardiovascular measurements in English with the Greek source	Counts and labels	Signed out → reference sections remain visible
B6	13:30–14:30	Crew Self	Explainable Alerts `/mission/alerts` and Health Intelligence `/mission/intelligence`	Open the crew evidence review; open evidence coverage	“Operational alerts: DISABLED”; observations or an empty state; coverage labelled “not medical findings”	Status texts	Empty → expected for accounts without data

STEP	TIME	ROLE	SCREEN	ACTION	EXPECTED RESULT	EVIDENCE	FAILURE RECOVERY
B7	14:30–15:30	Crew Self	Offline Mission Mode `/mission/offline` and Recorded Mission Data `/mission/records`	Read the local state; optionally switch to airplane mode and back; Sync now if earlier steps created records	Store READY; badge AUTHENTICATED or LOCAL ONLY; queue and storage columns	Status cards; storage filter	Sync locked → sign in again
B8	15:30–17:15	Mission Operator	Mission & Crew Operations `/mission/operations`	Select the synthetic mission; open the Synthetic mission demo journey; mark two tracker steps; choose a domain in the health preview; open the offline report and export JSON; export the integrated bundle and inspect it	Tracker counts update (session only); Day 0/7/14 table; local downloads; the bundle is accepted and an edited copy is rejected	Downloaded JSON; tracker text	Not an operator → ALT-OP
B9	17:15–18:45	Ground Medical	Ground & Research Console `/mission/ground`	Select the Ground Medical mission → Refresh pending reviews → View values. Record a decision only if the owner has authorized it for a newly submitted synthetic batch	“n pending synthetic batch(es).”; values table (metric, value, unit, excerpt); research-only notice; after an authorized decision, a receipt	Queue text; values table	No missions listed → ALT-GM
B10	18:45–20:00	Any	Wrap-up	Complete the evaluator checklist (Section 8.1)	Each checkpoint marked observed, blocked or not applicable	Your notes	—

6. Alternative Paths When a Role Is Not Available

Table 5. Alternatives (role-based access is never bypassed)

ID	Missing access	Alternative demonstration
ALT-CS	No Crew Self account	Show the B3 protocol library; open B4 without a Crew Self context to show the gated, local-only workbench; use the public B5 reference sections
ALT-OP	No Mission Operator account	Open `/mission/operations` to show the “Operator membership required” notice; describe the launchpad from Manual Section 13.3
ALT-GM	No Ground Medical account	Open `/mission/ground`: the review panel lists no missions and the database refuses calls — the isolation itself is the demonstration. Describe the decision flow from Manual Section 15.2
ALT-SENSOR	No phone or permission	Skip physical capture; point to DISCONNECTED cards and the local session context — evidence that nothing is fabricated
ALT-NET	NASA OSDR unavailable	Show the error state and that no substitute data is presented as live

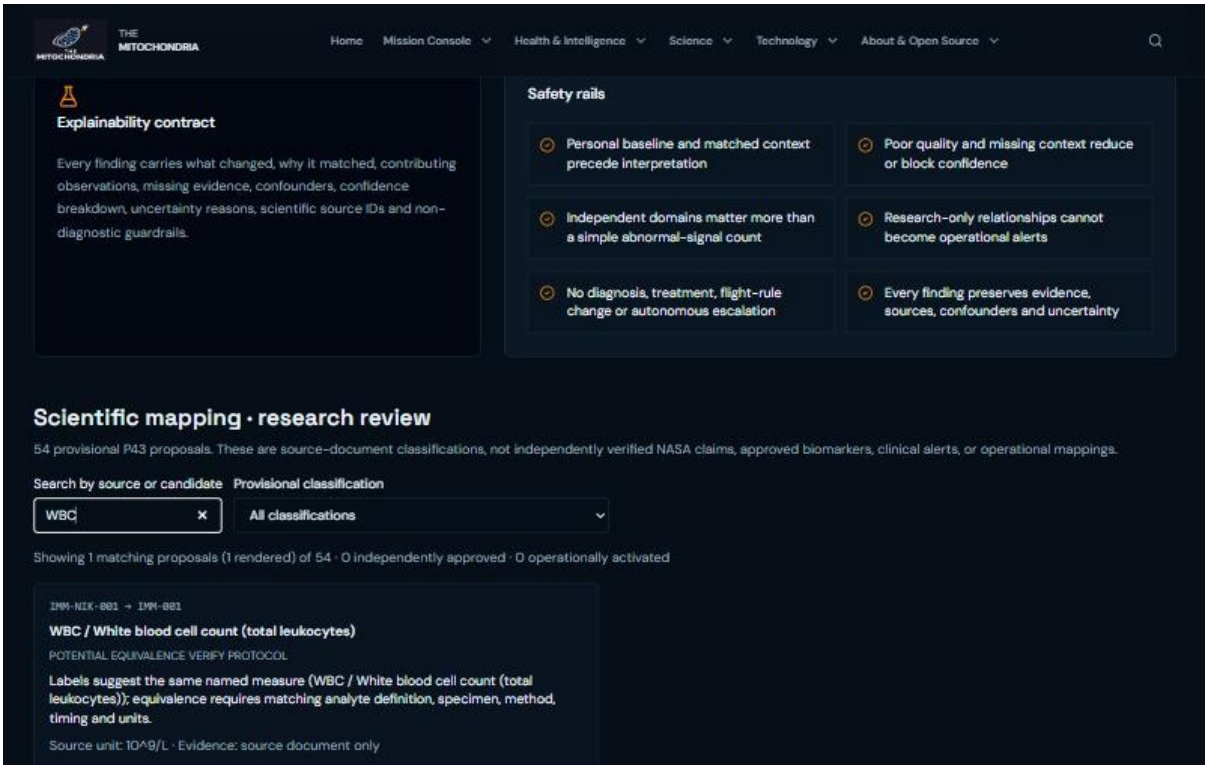
7. Synthetic versus Physical Evidence

Table 6. Evidence types

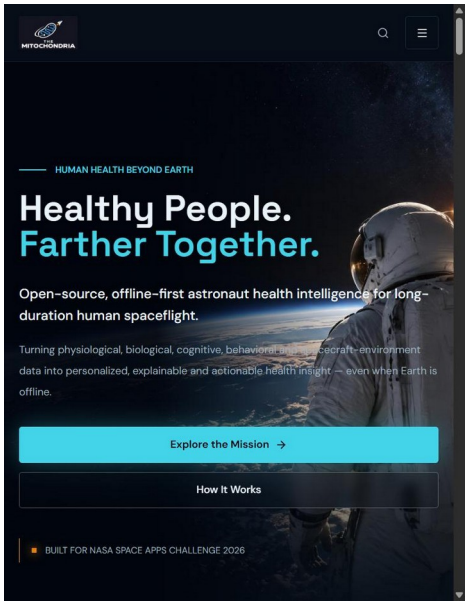
Evidence	Nature	How to describe it
Scenario fingerprints, demo dataset, offline report, integrated bundle	Synthetic	Software fixtures for demonstration and regression testing
Mission Documents candidates and Cloud batches	Synthetic values from a fictional report	Reviewed synthetic research data, not health records
Camera PPG and motion samples (optional)	Physical device input from the evaluator	Pulse estimate with quality; not ECG; not clinically validated; raw values are not exported
NASA OSDR metadata	Published research metadata (OSD-48 is mouse research)	Research-only, not astronaut data
Ground Medical decision receipt	Database correlation metadata	Research-only decision; not clinical approval

7.1 What the public path looks like

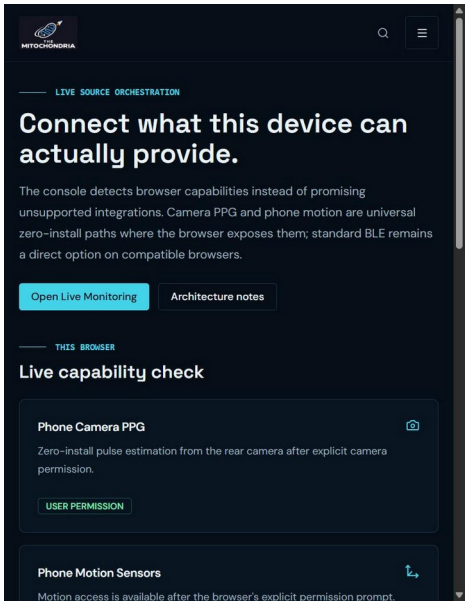
Unedited captures of the live application before sign-in: the Research Explorer on 23 October 2026 (desktop width) and the public pages on 18 October 2026 (765 × 990 px).



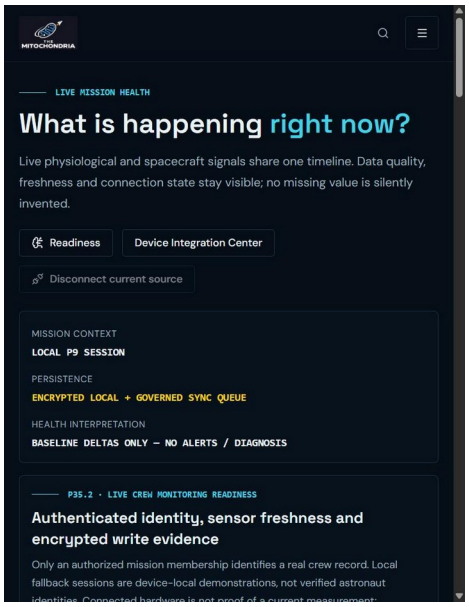
A6 · Health Intelligence Research Explorer — search “WBC” returns one provisional proposal (captured 23 October 2026)



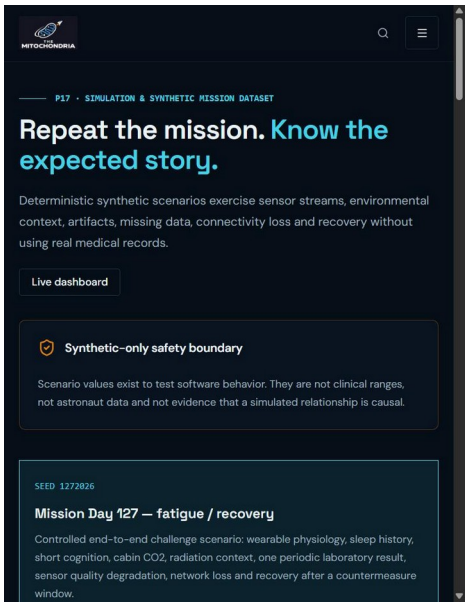
A1 · Home



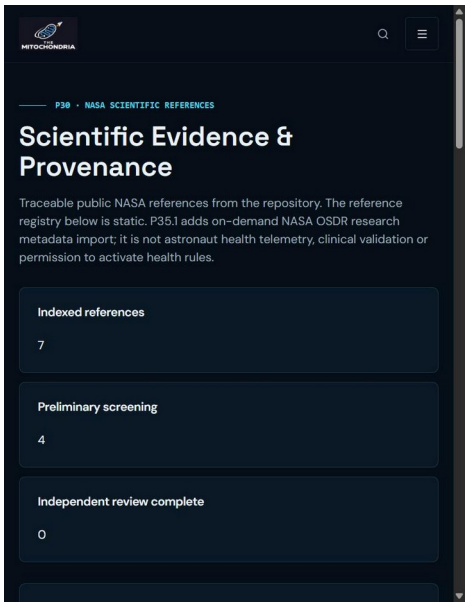
A2 · Device Integration Center



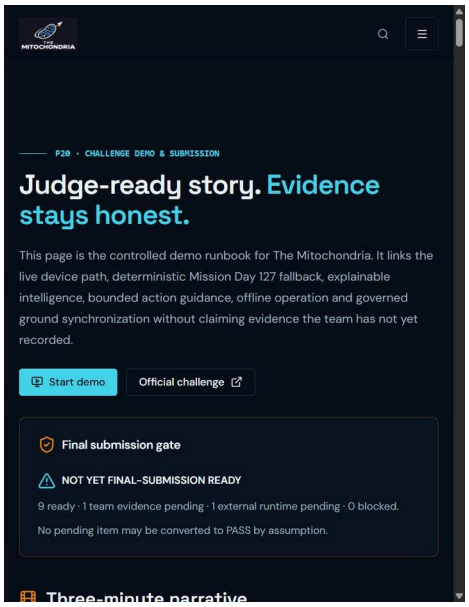
A3 · Live Monitoring (local session context)



A4 · Synthetic Mission Simulation



A5 · NASA Scientific Evidence



A6 · Challenge Demo & Submission

8. Troubleshooting and Evaluator Checklist

Problem	Fix
Sign-in error	Check the account e-mail; the generic message is intentional; try the e-mail magic link if the owner has enabled it
Wrong role shown	Select the correct membership card on /mission/identity
Launchpad controls disabled	The reserved acceptance mission is selected — choose your synthetic mission
Mission Documents finds nothing	Use the synthetic TXT fixture with exact catalogue names and units
Live identity shows LOCAL FALLBACK while signed in	Select Crew Self, then return focus to the tab (identity resolves on focus)
Camera fails	Allow permission; close other camera apps; use HTTPS — or skip (optional step)

8.1 Evaluator checklist

Table 7. Checklist

#	Checkpoint	Observed / Blocked / N/A
1	Public pages load and research-prototype disclaimers are visible	
2	No invented values when no source is connected	
3	Synthetic scenario is deterministic (same fingerprint)	
4	OSDR metadata is labelled research-only	
5	Research Explorer: WBC → 1, potential equivalences → 11, status line shows 0 approved and 0 activated	
6	Crew Self membership ACTIVE and MISSION-BOUND	
7	Self-assessment stored nothing	
8	Mission Documents accepted only exact matches and saved nothing without authorization	
9	Draft scientific catalogues are marked pending independent review	
10	Operator launchpad exports are local; the reserved acceptance mission is protected	
11	Ground Medical queue is visible only to Ground Medical; other roles see nothing	
12	No step bypassed role-based access	

9. Research Status at a Glance

Table 8. Status labels used across all three documents

Status	What it covers
OPERATIONAL / FIELD-VERIFIED	Identity and roles, mission operations and audit, live capture (camera and motion verified on a physical phone), encrypted offline store and sync, Mission Documents, synthetic Cloud batches, Ground Medical review
SYNTHETIC DEMONSTRATION	Scenarios, demo dataset, offline report, integrated bundle, synthetic laboratory reports

Status	What it covers
RESEARCH-ONLY	Health intelligence patterns, the Research Explorer of 54 provisional proposals (0 approved, 0 activated), domain evidence summaries, handbooks, draft catalogues, NASA OSDR metadata
VALIDATION BOUNDARY	Real-device offline OCR in airplane mode; two-account cross-mission Crew Self isolation; independent scientific review; live Ground Medical review acceptance and human real-device cases
OUT OF SCOPE	Persisted personal baselines, native Health Connect / HealthKit hosts, habitat telemetry feed, governed administrator role

- Not NASA-approved, not clinically validated, not flight-qualified; synthetic data only.
- Real-device evidence covers one phone (camera pulse estimate and motion); the Bluetooth path awaits a hardware test.
- Scientific content is research-only: Nikolas' immune and behavioural handbooks are integrated as 216 source records and 54 provisional proposals, all awaiting independent scientific review.
- Two-account cross-mission Crew Self isolation and airplane-mode offline OCR on a physical device have not been demonstrated.